Q09 Discussion Questions

1. FineWeb's global MinHash dedup removed 80% of the data and trained *worse* than per-snapshot dedup. When is a duplicate not a duplicate — that is, which kinds of repetition are noise (boilerplate) and which are signal (naturally recurrent facts, common code idioms)? What would a principled stopping rule for dedup aggressiveness look like?
2. FineWeb-Edu's classifier was trained on Llama-3 annotations that explicitly targeted grade-school-level educational value, and it still improved MMLU and ARC. What capabilities do you expect such a filter to *suppress*, and how would you detect that suppression in evaluation?
3. Muennighoff et al. find up to $\sim$4 epochs of repetition is nearly as good as fresh data, yet up-sampling a single domain 4$\times$ can change a model's behavior substantially. Is there a tension here, or are these simply different interventions (whole-corpus repetition vs. mixture reweighting)? Design the experiment that would separate them.
4. The tokenizer is trained on a sample of the mixture, and fertility differs by domain. Propose a concrete protocol for measuring how much of a declared mixture change is "real" (changed content) versus "artifactual" (changed token accounting), and predict where the artifact term is largest.
5. Given Villalobos et al.'s projection that the public human-text stock could be fully utilized between roughly 2026 and 2032, argue for or against: *data composition research is now more important to capability progress than architecture research.* What evidence over the next two years would change your mind?